

Prime Numbers as Emergent Sources in a Resonance Field

An exploratory dynamical reformulation

Status. This document is a concept note, not a proof of the Riemann hypothesis and not a claim that the exact harmonic identities below are new. Sections 2–6 formalize a known roots-of-unity divisibility filter and show that the resulting source rule is equivalent to a sieve. Sections 7–10 isolate the genuinely speculative question: whether an intrinsic operator or dynamical law can generate the source set without an externally imposed sequential scan and yield new analytic information.

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Abstract

This note develops a precise version of the intuition that prime numbers may be viewed as newly born sources in a field of periodic resonances. A source labelled by an integer p emits a roots-of-unity wave whose intensity at an integer n is one exactly when $p \mid n$ and zero otherwise. A candidate becomes a new source if no earlier source resonates at that site. The exact recursive rule generates precisely the primes, and prime-power resonance towers recover the valuations $v_p(n)$ and hence unique factorization. This formal part is a harmonic restatement of divisibility and sieving. The proposed research direction is to replace the explicit sequential rule by an autonomous field, operator, transfer mechanism, or variational principle whose spectrum or source-birth events recover primes and, ideally, connect local divisibility waves with the global oscillations arising from the non-trivial zeros of the Riemann zeta function. Concrete criteria for novelty and questions for specialists are stated.

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1 Motivation and scope

The basic picture is the following. Integer sites form a one-dimensional medium. Once a site p is declared an active source, it generates a periodic resonance at

$$p, 2p, 3p, \dots$$

A composite integer is a site reached by at least one previously active source. A prime is a site not reached by any relevant earlier source; after being found, it becomes a source itself.

This language is deliberately physical, but it can be made exact. The exact model below does *not* yet improve on the sieve of Eratosthenes. Its purpose is to identify a mathematically clean state space from which one can ask whether a more intrinsic dynamical or spectral construction exists.

2 An exact harmonic divisibility wave

For an integer $q \geq 2$, define the complex periodic kernel

$$D_q(x) = \frac{1}{q} \sum_{r=0}^{q-1} \exp\left(\frac{2\pi i r x}{q}\right), \quad x \in \mathbb{R}. \quad (1)$$

Its measurable “wave intensity” is $I_q(x) = |D_q(x)|^2$.

Lemma 1 (Roots-of-unity filter). *For every integer n ,*

$$D_q(n) = I_q(n) = \begin{cases} 1, & q \mid n, \\ 0, & q \nmid n. \end{cases} \quad (2)$$

Proof. If $q \mid n$, every summand in (1) equals 1. Otherwise the summands form a non-trivial geometric progression whose ratio is a q th root of unity different from 1, so their sum is zero. $\square$

Away from removable singularities, the intensity also has the real form

$$I_q(x) = \frac{1}{q^2} \left(\frac{\sin(\pi x)}{\sin(\pi x/q)} \right)^2. \quad (3)$$

Thus I_q is a continuous periodic interference pattern, while at integer sites it becomes an exact divisibility detector.

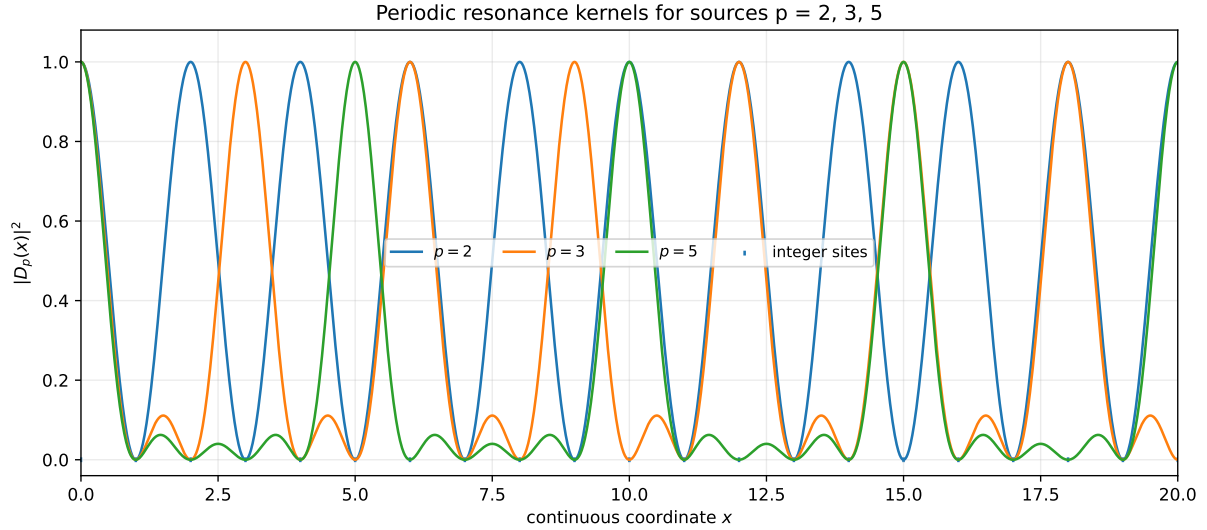


Figure 1: Continuous resonance intensities. At integer sites, the p -kernel is 1 exactly at multiples of p and 0 at all other integers.

3 The recursive source-birth rule

Let $\mathcal{P}_{< n}$ denote the set of sources born before the candidate site n . Define the local resonance energy

$$E(n) = \sum_{\substack{p \in \mathcal{P}_{< n} \\ p \leq \sqrt{n}}} I_p(n). \quad (4)$$

The source-birth indicator is

$$B(n) = \mathbf{1}_{\{E(n)=0\}}. \quad (5)$$

Starting with $\mathcal{P}_{< 2} = \emptyset$, scan $n = 2, 3, 4, \dots$ and add n to the source set exactly when $B(n) = 1$.

Proposition 1. *The source set generated by (4)–(5) is exactly the set of prime numbers.*

Proof. Assume all sources below n are exactly the primes below n . If n is composite, it has a prime divisor $p \leq \sqrt{n}$. That p is already active, $I_p(n) = 1$, and therefore $E(n) > 0$ and $B(n) = 0$. Conversely, if $E(n) = 0$, no prime $p \leq \sqrt{n}$ divides n , so n is prime. The statement follows by induction. $\square$

The proposition is exact, but it also exposes the present limitation: the model uses the same causal order and the same divisibility information as a classical sieve.

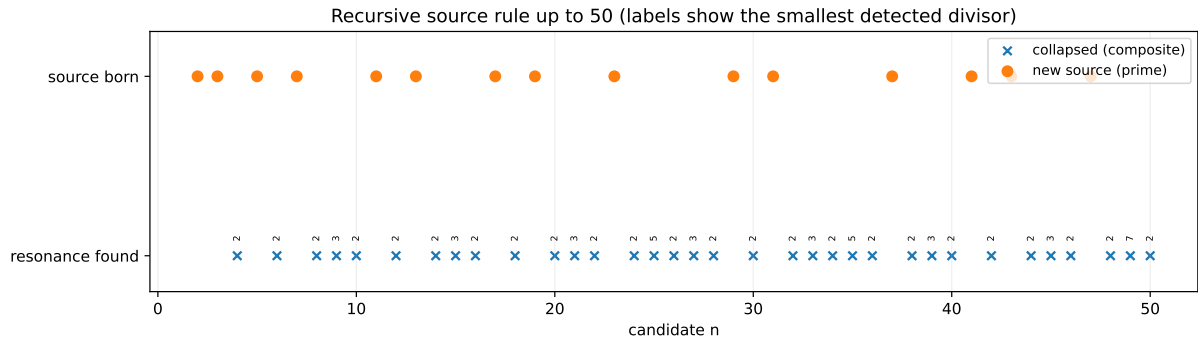


Figure 2: Source births and collapses up to 50. A prime is a candidate for which no active source with $p \leq \sqrt{n}$ resonates.

3.1 Small numerical example

n	Earlier resonant sources $p \leq \sqrt{n}$	Event
2	none	source born
3	none	source born
4	2	collapse: $4 = 2^2$
5	none	source born
6	2	collapse: $6 = 2 \cdot 3$
7	none	source born
8	2	collapse
9	3	collapse: $9 = 3^2$
10	2	collapse
11	none	source born
12	2, 3	multiple resonance
13	none	source born
14	2	collapse
15	3	collapse
16	2	collapse
17	none	source born
18	2, 3	multiple resonance

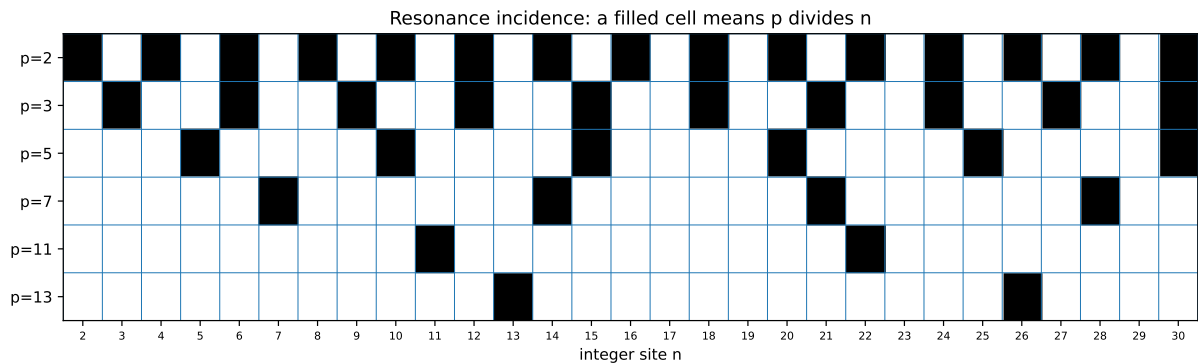


Figure 3: Incidence matrix for several prime sources. Intersections of rows represent integers with several distinct prime factors.

4 Prime-power towers and exact factorization

The first-level resonance $I_p(n)$ records whether p divides n , but not its multiplicity. Introduce the tower

$$V_p(n) = \sum_{k \geq 1} I_{p^k}(n). \quad (6)$$

Only finitely many terms are non-zero, and

$$V_p(n) = v_p(n), \quad (7)$$

where $v_p(n)$ is the usual p -adic valuation. Consequently,

$$n = \prod_p p^{V_p(n)}. \quad (8)$$

The entire factorization of n is therefore its vector of tower intensities.

For example,

$$72 = 2^3 3^2, \quad V_2(72) = 3, \quad V_3(72) = 2,$$

and all other $V_p(72)$ vanish. With logarithmic source weights, unique factorization becomes an additive energy identity:

$$\log n = \sum_p V_p(n) \log p. \quad (9)$$

Other arithmetic observables can also be written in this language. For example,

$$\omega(n) = \sum_p I_p(n)$$

is the number of distinct prime factors. Squarefreeness is equivalent to $\sum_p I_{p^2}(n) = 0$.

5 Euler product as a partition function of source towers

For $\Re(s) > 1$,

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}} = \prod_p (1 + p^{-s} + p^{-2s} + \dots) \quad (10)$$

$$= \prod_p \sum_{k \geq 0} \exp(-sk \log p). \quad (11)$$

Each prime source contributes an independent tower with levels $k \log p$. Multiplying all towers produces each integer exactly once because the exponent vector $(v_p(n))_p$ is unique.

The logarithmic derivative isolates prime-power emissions:

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n \geq 1} \frac{\Lambda(n)}{n^s} = \sum_p \sum_{k \geq 1} (\log p) e^{-sk \log p}. \quad (12)$$

This makes the source-tower interpretation more than visual: the standard analytic object that connects primes to the zeros of ζ is already a weighted superposition of prime-power levels.

6 Local resonance waves versus zeta zeros

It is essential not to identify two different kinds of oscillation.

1. The kernels $D_p(x)$ encode *local periodic divisibility* on the additive integer axis.
2. The non-trivial zeros $\rho = \beta + i\gamma$ generate *global logarithmic oscillations* in explicit formulas for prime-counting functions.

For example, with the usual symmetric interpretation of the zero sum and the standard half-weight convention at discontinuities, an explicit formula has the schematic form

$$\psi_0(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2} \log(1 - x^{-2}). \quad (13)$$

The imaginary parts γ act as frequencies in the variable $\log x$. The local source field and the global zero field contain related arithmetic information, but the derivation of one from an autonomous dynamics of the other is exactly the non-trivial issue.

7 A speculative continuous source-collapse dynamics

The exact model still performs an ordered scan. To formulate a research target, replace the sharp comb by a smoothed periodic field, for example

$$K_{p,\sigma}(x) = \sum_{m \geq 2} \exp\left(-\frac{(x - mp)^2}{2\sigma^2}\right), \quad (14)$$

and let the active-source amplitude at integer site n be $a_n(t) \in [0, 1]$. A generic field is

$$\mathcal{E}_{\sigma}(x, t) = \sum_{m \geq 2} a_m(t) K_{m,\sigma}(x). \quad (15)$$

One possible thresholded birth law is

$$\dot{a}_n(t) = \chi_n(t)(1 - a_n(t))F_{\beta}(\theta - \mathcal{E}_{\sigma}(n, t)), \quad F_{\beta}(y) = \frac{1}{1 + e^{-\beta y}}. \quad (16)$$

Here $\chi_n(t)$ is an eligibility or causal-front term. In a singular limit ($\sigma \rightarrow 0$, $\beta \rightarrow \infty$) and with eligibility imposed in increasing order, this recovers a softened version of the sieve.

The key weakness is also visible: an externally prescribed χ_n simply hides the sequential scan. A meaningful advance would require an intrinsic propagation law, conservation principle, variational condition, or operator evolution that creates the same source set without explicitly inserting primality, divisibility, or the order 2, 3, 4, ... into the rule.

8 What would constitute a genuinely new contribution?

A source-resonance model would become mathematically substantive if it achieved at least one of the following.

1. **Autonomous generation.** Produce the prime source set from a fixed operator or evolution law without candidate-by-candidate trial division.

2. **New quantitative information.** Prove a bound on prime gaps, the error in the prime number theorem, correlations of Λ , or the location/statistics of zeta zeros that is not a disguised classical argument.
3. **Spectral determinant.** Construct a natural operator L for which a determinant, trace formula, or periodic-orbit expansion yields $\zeta(s)$, $1/\zeta(s)$, or an equivalent completed zeta function.
4. **Natural continuum limit.** Derive the density $1/\log x$ and its fluctuations from the dynamics, rather than inserting it.
5. **Robustness and universality.** Explain why perturbations of the field preserve statistical laws associated with primes or with random-matrix predictions for zeta zeros.
6. **Computational leverage.** Obtain a provably different complexity profile from existing sieves or primality tests.

Without such an outcome, the construction remains a useful visualization and a harmonic encoding of known arithmetic.

9 Concrete research questions for a specialist

1. Is there an established framework—dynamical zeta functions, transfer operators, trace formulas, arithmetic quantum chaos, or noncommutative geometry—in which the recursive source-birth rule has a natural analogue?
2. Can the roots-of-unity field D_p be assembled into a single operator whose spectrum separates primes from composites without sequentially testing divisors?
3. Is there a natural determinant identity connecting such an operator to the Euler product?
4. Can the local resonances at multiples mp be interpreted as periodic orbits whose lengths are $\log p$, thereby relating the construction to explicit formulas and Hilbert-Pólya-type ideas?
5. Does a smoothed field admit a continuum or renormalization limit from which $\pi(x) \sim x/\log x$ follows?
6. Can the interaction energy be chosen so that $\Lambda(n)$ or $\mu(n)$ emerges as an observable, rather than being encoded by hand?
7. Is there any obstruction theorem showing that an autonomous local dynamics of this type cannot generate primes more efficiently than a sieve?
8. What is the closest known literature, and which part of this formulation is merely a change of language?

10 Suggested computational experiments

The following experiments would not prove novelty, but could clarify the structure.

1. Compare hard kernels I_p and smoothed kernels $K_{p,\sigma}$; measure false births as functions of σ and threshold θ .
2. Introduce finite propagation speed and delayed source activation; determine whether the correct source set is stable under timing perturbations.
3. Study spectra of finite matrices

$$M_{p,n} = I_p(n), \quad p \leq P, \quad 2 \leq n \leq N,$$

- and of weighted Gram matrices built from their rows.
4. Test whether singular vectors or eigenvalue statistics of these incidence operators correlate with prime gaps or with values of arithmetic functions.
 5. Search for a transfer operator whose periodic-orbit expansion produces the weights $\log p$ appearing in (12).

11 Conclusion

The exact statement can be summarized in one sentence:

A prime is a site at which all earlier relevant divisibility waves destructively cancel; once identified, that site becomes a new periodic source.

Mathematically, this is an exact roots-of-unity reformulation of sieving and unique factorization. Its potential lies not in that reformulation alone, but in the possibility—still entirely open in this note—that the source set might arise as the spectrum, orbit structure, or stable state of a natural autonomous system. The central task is therefore not to add more wave metaphors, but to identify a single non-circular operator or dynamical law and extract a theorem from it.

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